Exercise 4

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Data Science class

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Joins, Union, Filtering, and Aggregates

1. Select A.EmployeeID, A.firstName, A.LastName, A.Department
A.salary, B.ProjectID, B.ProjectName, B.Budget
B.status,
from Employees as A
inner join Projects A B
on A.EmployeeID = B.EmployeeID;

Emp ID	first name	Last name	Depart-Ment	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Dev	100 000	complete
1	John	Doe	IT	70 000	103	Cyber Security Aud	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	finance	75 000	104	Financial Analysis	90 000	ongoing
5	Emma	wilson	sales	65 000	105	market Expansion	65 000	Completed
6	micheal	clark	finance	80 000	106	Risk Management	80 000	Pending

2. Select A.EmployeeID, A.firstname, A.Lastname, A.department
A.salary, B.ProjectID, B.Projectname, B.Budget
B.status
from Employees As A
Left join Projects As B
on A.EmployeeID = B.EmployeeID

Emp ID	First name	Last name	Department	Salary	Project ID	project name	Budget	status
1	John	Doe	IT	70 000	101	AI DEV	100000	Completed
1	John	Doe	IT	70 000	103	cybersec	75 000	Pending
2	Alice	Smith	HR	60 000	102	Emp Trai	50000	ongoing
3	Bob	Johnson	Finance	75 000	104	Fin Analysis	90 000	ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market exp	65 000	completed
6	Micheal	Clark	Finance	80 000	106	Rist Manag	80 000	Pending

3. Select B.Project ID, B.Project_name, B.Budget, B.status, A.EmployeeID, A.Firstname, A.Lastname, A.department, A.Salary
From employees as A
Right Join projects as B
On A.employee ID = B.employee ID

Project ID	Project Name	Budget	status	Employee ID	First name	Last name	department	Salary
101	AI Development	100 000	completed	1	John	Doe	IT	70 000
102	Employee Traing	50 000	ongoing	2	Alice	Smith	HR	60 000
103	Cybersec Audit	75000	Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90 000	Ongoing	3	Bob	Johnson	Finance	75 000
105	market expansion	65 000	completed	5	Emma	Wilson	Sales	65 000
106	Risk management	80 000	Pending	6	Micheal	Clark	Finance	80 000

4. Select A.employee ID, A.firstname, A.Lastname, A.department, A.Salary, B.Project ID, B.projectname, B.budget, B.status

from employees as A
full outer join projects as B.
on A.employeeID = B.employeeID;

Emp ID	First name	Last name	depart-ment	salary	Project ID	Project name	Budget	status
1	John	Doe	IT	70000	101	AI Developm	100000	completed
1	John	Doe	IT	70000	103	Cyber security	75000	pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	ongoing
4	David	Brown	IT	72000	~~Null~~ ~~105~~	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	market expansion	65000	completed
6	Micheal	Clark	Finance	80000	106	Risk Managem.	80000	Pending

5. Select city As Location
from employees
Union
Select status As Location
from projects;

Location
New York
Los Angeles
Toronto
London
Sydney
completed
ongoing
Pending

6. Select city As Location
from employees
Union all
Select status As Location
from projects;

Location
New york
Los Angeles
Toronto
London
Sydney
New york
completed
Ongoing
Pending
Ongoing
Completed
Pending

7. Select EmployeeID, Firstname, Lastname, department, Salary
From employees
Where salary > 70 000;

EmployeeID.	firstname	Lastname	department	Salary
~~1~~	~~John~~	~~Doe~~	~~IT~~	~~80 000~~
3	Bob	Johnson	Finance	95 000
4	David	Brown	IT	72 000
6	Michael	clark	Finance	80 000

8. Select employeeID, firstname, Lastname, department, salary
from employees
where department in ('IT', 'finance');

EmployeeID	firstname	Lastname	department	salary
1	John	Doe	IT	70000
3	Bob	Johnson	finance	75000
4	David	Braun	IT	72000
6	Michael	clark	finance	8000

9. Select ProjectID, projectname, Budget, status
from projects
where status <> 'completed';

ProjectID	Projectname	Budget	status
102	Employee Train	50 000	ongoing
103	Cybersec Audit	75 000	pending
104	financial analysis	90 000	ongoing
106	Risk management	80 000	pending

10. Select ProjectID, projectname, budget, status
from projects
where budget > 70000
and status <> 'completed';

Project ID	Projectname	Budget	status
103	Cybersec Audit	75000	pending
104	Financial analysis	90 000	ongoing
106	Risk management	80 000	pending

11. Select employeeID, Firstname, Lastname, department, salary, City
from employees
where city in ('New york', 'Toronto')
Order by Salary Desc;

EmployeeID	Firstname,	Lastname	department	Salary	city
6	Michael	Clark	Finance	80000	NY
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	NY

12. Select employeeID, Firstname, Lastname, department, Salary
from employees
Order by salary Desc
Limit 3;

EmployeeID	Firstname	Lastname	department	salary.
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

13. Select department, sum(salary) As total salary
from employees
Group by department
Order by total salary Desc;

Department	total_Salary
~~IT~~ Finance	155 000
~~Finance~~ IT	142 000
~~HR~~ Sales	65 000
HR	60 000

14. Select City, Avg (Salary) as average_salary
from employees
Group by city
Having Avg (salary) > 65000;

City	average_salary
New York	75 000
Toronto	75 000
London	72 000

15. Select department, count (employeeID) As employee_count
from employees
Group by department
Having count (employeeID) > 1;

Department	employee_count
IT	2
Finance	2

16. Select status, count (ProjectID) As Project_count
from projects
Group by status
Having count (Project ID) >= 2;

Status	Project-Count
Completed	2
Ongoing	2
Pending	2

17- Select A.employeeID, A.first_name, A.last_name, sum (B.budget) As total_project_budget
from employees As A
inner join projects As B
on A.EmployeeID = B.employee ID
Group by A.employeeID, A.firtname, A.lastname
Having sum (B.budget) > 150000;

EmployeeID	firstname	Lastname	Total_project_budget
1	John	Does	175 000